# Model card — Random Forest supplied-label classifier

## Model identity

| Field | Recorded value |
|---|---|
| Model | Random Forest |
| Task | Binary classification of supplied `has_risk` labels |
| Primary tuning signal | Training grouped-CV average precision |
| Internal split | 12,459 train / 3,115 test; 0 formula-group overlap |
| Operating threshold | 0.438665; chosen from training OOF predictions |
| Masking | Not applied ( `masking_applied: false` ) |

## Intended use

This is an educational screening prototype for reproducing a supplied historical label from a product's ingredient list. It may support exploration of model behavior in this dataset. It is not suitable for medical, toxicological, consumer-safety, regulatory, compliance, or individual health decisions.

## Training data and provenance

The supplied labeled input has 15,621 rows. Reproducible cleaning quarantines 47, leaving 15,574 products (4,059 positive; 11,515 negative). The exact registry behind the stated 16-substance count and the original labeling code are unavailable. The target therefore remains a supplied proxy label, not independently verified substance presence or safety.

Formula grouping uses normalized, order-insensitive ingredient sets. It prevents exact normalized formula groups from crossing the internal split but does not prove that nearly identical products, population shifts, or external products will generalize equally well.

# Inputs and feature construction

The Random Forest uses top-1,500 **training-fold ingredient-presence** binary columns and **`log1p(ingredient_count)`**. It does not use categories, brands, product names, concentrations, country tags, external reference matches, or an inferred list of 16 substances.

# Selection and tuning record

The model is the final ensemble-focused pipeline. The benchmark CV leader was **Balanced Logistic Regression**; this field can differ from the selected Random Forest and is retained rather than hidden. Random Forest tuning uses training-only grouped CV average precision.

- Training grouped-CV average precision: **0.9990**.
- Training grouped-CV F1: **99.86%**.

| Parameter | Recorded value |
|---|---|
| `class_weight` | `balanced_subsample` |
| `max_features` | `0.2` |
| `min_samples_leaf` | `2` |
| `n_estimators` | `200` |

# Internal hold-out performance

## Default 0.5 threshold

| Metric | Value |
|---|---|
| Accuracy | **99.90%** |
| Precision | **100.00%** |
| Recall | **99.63%** |
| F1 | **99.81%** |
| F2 | **99.70%** |
| ROC-AUC | **0.9990** |
| Average precision | **0.9985** |

### Training-selected threshold (0.438665)

| Metric | Value |
|---|---|
| Accuracy | **99.90%** |
| Precision | **100.00%** |
| Recall | **99.63%** |
| F1 | **99.81%** |
| F2 | **99.70%** |
| ROC-AUC | **0.9990** |
| Average precision | **0.9985** |

Rows are supplied labels; columns are model predictions.

| | Predicted 0 | Predicted 1 |
|---|---|---|
| **Observed 0** | 2,303 | 0 |
| **Observed 1** | 3 | 809 |

These results are agreement metrics against supplied labels on an internal grouped holdout. They are not external validation, calibration evidence, or evidence of toxicity prediction.

## Recognized-scope sensitivity

The operational taxonomy-recognized scope has **6,785 rows**. **Recognized-category sensitivity reuses/overlaps the primary source cohort and is not independent external validation; metrics use default threshold 0.5.**

| Metric | Mean | Fold standard deviation |
|---|---|---|
| Accuracy | **99.79%** | 0.23% |
| Precision | **100.00%** | 0.00% |
| Recall | **99.21%** | 0.90% |

| | | |
|---|---|---|
| F1 | **99.60%** | 0.46% |
| F2 | **99.36%** | 0.72% |
| ROC-AUC | **0.9991** | 0.0007 |
| Average precision | **0.9986** | 0.0013 |

## Explanation and limits

MDI importance values are non-negative. They should not be interpreted as direction of risk, signed benefit/harm, causal effect, ingredient toxicity, or safety certification. The application provides local **drop-one ingredient probability perturbations**, not SHAP explanations. A probability is a model score for the supplied target, not a calibrated personal-risk estimate.

No hosting, OCR, complete regulatory taxonomy, full synonym harmonization, deterministic 16-item matching evaluation, or independent label verification is included.

## Evidence and maintenance

- Machine-readable metrics
- Readable final-results report
- Test output: 23 passed
- Notebook execution summary: 43 notebook cells, 24 executed code cells, 0 error outputs, 15 embedded PNG figures.

Only load model artifacts from trusted sources and regenerate them with compatible package versions when needed.